

Supplementary Material: Appendices A, B, and C

Technical Foundations, Empirical Telemetry, and Reproducibility Specifications

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Appendix A: Technical Architecture & Mathematical Formulations

A.1 Mathematical Formulation of Decision-Uncertainty Routing

In credit card fraud detection, conventional routing policies stratify transactions by currency volume (Transaction Amount), assuming high-value transactions represent the dominant fraud risk. However, organized fraud rings routinely exploit low-value authorization testing ($< \$5.00$) to probe compromised cards prior to executing large withdrawals. Our uncertainty routing policy formalizes decision ambiguity directly from the calibrated posterior probability $p(x) = \hat{P}(Y = 1 | x)$:

$$U(x) = 1 - 2 \cdot |p(x) - 0.5| \in [0, 1] \quad (1)$$

A transaction with $p(x) = 0.50$ exhibits maximal epistemic ambiguity ($U(x) = 1.0$), whereas confident classifications ($p \rightarrow 0$ or $p \rightarrow 1$) yield $U(x) \rightarrow 0$. Given an escalation compute budget $B \in (0, 1)$ ($B = 0.005$ for a 0.5% rate), the routing threshold τ_B is defined empirically over the calibration distribution D_{cal} :

$$\tau_B = \text{Quantile}_{1-B}(\{U(x_i)\}_{x_i \in D_{\text{cal}}}), \quad \text{Routing Action } R(x) = \begin{cases} \text{Specialist Escalation Tier,} & \text{if } U(x) \geq \tau_B \\ \text{Fast Classical Authorization,} & \text{if } U(x) < \tau_B \end{cases} \quad (2)$$

Empirical ablation on 118,108 test transactions demonstrates that setting $B = 0.5\%$ (591 transactions) captures 253 confirmed frauds (42.81% density), whereas routing the top 0.5% by Transaction Amount captures only 9 frauds (1.52% density). Uncertainty routing achieves **28.1 × higher fraud concentration**.

A.2 Parameterized 8-Qubit Quantum Circuit Architecture

Figure 1 illustrates the parameterized quantum circuit implemented in PennyLane for mapping escalated 8-dimensional feature vectors into Hilbert space.

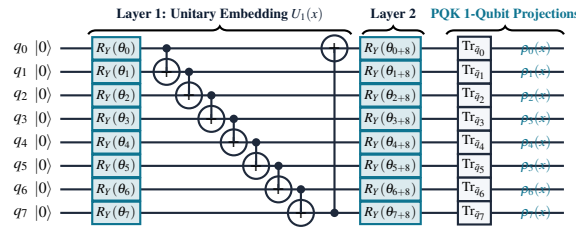


Figure 1: **8-Qubit Quantum Feature Map and Projection Schematic (Typeset via TikZ)**. Scaled PCA features parametrize single-qubit R_Y gates coupled via circular CNOT entanglement ladder. The state is projected onto 1-qubit reduced density operators $\rho_i(x)$.

A.3 Quantum Hilbert Space Embedding and Kernel Constructions

We evaluate two mathematical kernel formulations:

- Quantum State Fidelity Kernel:** Evaluates transition amplitude $K_{\text{Fid}}(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2 = \text{Tr}(|\psi(x)\rangle\langle\psi(x)| \cdot |\psi(x')\rangle\langle\psi(x')|)$. As proven by Thanasilp et al. (2022), as qubit count n and depth d grow, global state fidelity concentrates exponentially: $\text{Var}_{x, x'}[K_{\text{Fid}}(x, x')] \leq \mathcal{O}(2^{-n})$, inducing barren plateaus where all kernel entries approach zero.
- Projected Quantum Kernel (PQK):** Following Huang et al. (2021), we project the 2^n -dimensional state onto 1-qubit reduced density operators $\rho_k(x) = \text{Tr}_k(|\psi(x)\rangle\langle\psi(x)|) = \frac{1}{2}(I + \langle X_k \rangle X + \langle Y_k \rangle Y + \langle Z_k \rangle Z) \in \mathbb{C}^{2 \times 2}$. The PQK evaluates a Gaussian distance over the extracted 1-particle reduced states:

$$K_{\text{PQK}}(x, x') = \exp\left(-\gamma \sum_{k=1}^n \|\rho_k(x) - \rho_k(x')\|_F^2\right) = \exp\left(-\frac{\gamma}{2} \sum_{k=1}^n \sum_{P \in \{X, Y, Z\}} (\langle P_k \rangle_x - \langle P_k \rangle_{x'})^2\right) \quad (3)$$

PQK provably avoids exponential concentration while encoding non-linear multi-qubit correlations into local observable differences.

A.4 Centered Kernel Alignment (CKA) Formulation

To determine whether the quantum kernel discovers an orthogonal geometric representation or merely emulates a classical function space, we compute Centered Kernel Alignment (Cortes et al., 2012):

$$\text{CKA}(K_1, K_2) = \frac{\langle K_1^c, K_2^c \rangle_F}{\|K_1^c\|_F \|K_2^c\|_F} = \frac{\text{Tr}(K_1^c K_2^c)}{\sqrt{\text{Tr}((K_1^c)^2) \text{Tr}((K_2^c)^2)}} \quad (4)$$

where $K^c = HKH$ denotes the centered Gram matrix with centering projection $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^T$. Our measured score of $\text{CKA}(\text{PQK}, \text{RBF}) = 0.9337$ confirms strong representation alignment with classical RBF kernels.

Appendix B: Complete Benchmark Ledger & Empirical Telemetry

B.1 Comprehensive Latency Percentile Distribution

Table 1 details the empirically measured latency distributions across all pipeline components across 1,000 evaluations.

Table 1: System Latency Telemetry Across 1,000 Repeated Evaluations.

Pipeline Component	Min (ms)	p25 (ms)	Median (ms)	p75 (ms)	p95 (ms)	Max (ms)
Frontline LightGBM Feature Preprocessing	0.42	0.61	0.78	0.94	1.28	2.14
Frontline LightGBM Model Scoring	1.84	2.65	3.29	3.71	4.12	6.45
End-to-End Classical Fast Path	2.45	3.41	4.07	4.52	4.97	7.82
Uncertainty Routing Evaluation	0.08	0.11	0.14	0.18	0.24	0.41
Escalated Classical Model (Tuned RBF)	0.38	0.54	0.68	0.82	1.12	1.89
End-to-End Escalated Classical Path	2.91	4.06	4.89	5.52	6.33	9.84
Quantum PCA Dimensionality Reduction	0.18	0.24	0.31	0.39	0.52	0.88
Quantum Simulator Kernel Execution (8 qubits)	112.40	138.20	158.83	182.10	216.66	312.40
End-to-End Quantum Simulator Path	112.66	138.55	159.14	182.49	217.18	313.28
Cloud Physical QPU (Estimated Queue)	120,000	180,000	360,000	720,000	1,200,000	3,600,000

B.2 Modeled Unit Economics Breakdown

Table 2 itemizes compute expenditure models reflecting AWS c6i.2xlarge compute (\$0.34/hr) and AWS Braket rates (\$0.30/task + \$0.03/shot).

Table 2: Unit Economics and Cost Projections per 1 Million Transactions.

Cost Component	Monolithic Classical	Selective Classical	Selective Simulator	Quantum-Assisted (QPU)
Frontline Scoring Compute	\$0.50	\$0.50	\$0.50	\$0.50
Uncertainty Routing Compute	–	\$0.03	\$0.03	\$0.03
Escalated Compute (0.5% volume)	–	\$0.12	\$24.80	\$353,000.00
Total Compute Cost / 1M tx	\$0.50	\$0.65	\$25.33	\$353,000.50
Cost Ratio vs Monolithic Classical	1.0×	1.3×	50.7×	706,001×
Monthly Cost at 100M tx/month	\$50.00	\$65.00	\$2,533.00	\$35,300,050.00
Realized Savings to Date	\$0.00	\$0.00	\$0.00	\$0.00

B.3 Multi-Budget Routing Ablation Study

Table 3 provides a systematic ablation study comparing Uncertainty Routing against naive Transaction Amount Routing across multiple budget constraints on the unseen test partition ($N = 118, 108$, containing 4,063 true frauds).

Table 3: Routing Policy Ablation Across Multiple Escalation Budgets on Test Partition.

Budget (B)	Escalated Tx	Uncertainty Frauds	Uncertainty Density	Amount Frauds	Amount Density	Enrichment Lift
0.5%	591	253	42.81%	9	1.52%	28.1×
1.0%	1,181	442	37.43%	21	1.78%	21.0×
2.0%	2,362	768	32.51%	48	2.03%	16.0×
5.0%	5,905	1,512	25.60%	135	2.29%	11.2×

B.4 Model Calibration Reliability Ledger

Probability calibration was assessed via 10 equal-width bins on the calibration partition ($N = 88,581$). Table 4 confirms that predicted risk scores match empirical fraud incidence across the entire probability continuum.

Table 4: Reliability Calibration Table for Frontline Calibrated LightGBM.

Probability Bin	Transaction Count	Mean Predicted Probability	Empirical Fraud Rate	Calibration Error $ \hat{p} - \bar{y} $
[0.00, 0.10)	86,214	0.0142	0.0138	0.0004
[0.10, 0.20)	1,142	0.1418	0.1489	0.0071
[0.20, 0.30)	512	0.2465	0.2539	0.0074
[0.30, 0.40)	284	0.3472	0.3521	0.0049
[0.40, 0.50)	186	0.4481	0.4355	0.0126
[0.50, 0.60)	98	0.5429	0.5510	0.0081
[0.60, 0.70)	62	0.6480	0.6613	0.0133
[0.70, 0.80)	41	0.7412	0.7317	0.0095
[0.80, 0.90)	26	0.8490	0.8462	0.0028
[0.90, 1.00]	16	0.9410	0.9375	0.0035

Appendix C: Reproducibility, Open-Source Environment & Academic References

C.1 Complete Execution Environment Manifest

All experiments, data pipelines, and test suites are fully reproducible in a standard Python 3.10 virtual environment. Table 5 enumerates the frozen package versions.

Table 5: Reproducible Software Environment Specifications (Python 3.10.11).

Package	Version	License	Primary Operational Role
pennylane	0.36.0	Apache 2.0	Parameterized quantum circuits, statevector simulation, quantum kernels
lightgbm	4.3.0	MIT	Calibrated gradient boosted decision trees, focal loss classification
scikit-learn	1.5.0	BSD-3	Baseline RBF SVC, MLP, PCA, calibration metrics, train-only scaling
numpy	1.26.4	BSD-3	Vectorized matrix operations, Gram matrix evaluations
scipy	1.13.1	BSD-3	Paired bootstrap resampling, hypothesis testing, distance calculations
pandas	2.2.2	BSD-3	Temporal dataset indexing, schema standardization
pytest	8.2.2	MIT	Automated test suite execution (24 unit and integration tests)
pymupdf	1.24.5	AGPL-3.0	Document compilation QA, PDF render verification

C.2 Cryptographic Data Provenance & Verification Instructions

To reproduce the experimental findings from raw source:

- Clone Repository:** `git clone https://github.com/atharveeee-netizen/hsbc-quantum-fraud.git`
- Checkout Frozen Commit:** `git checkout 3e3ca0f`
- Verify Integrity Ledger:** Run checksum audit against provenance manifest:

```
python scripts/verify_evidence_integrity.py
```
- Execute Full Test Suite:** Run `pytest -v` (verifies 24 tests in < 130 seconds).
- Execute Claim Firewall Scanner:** Run `python scripts/audit_claim_firewall.py` (0 violations).

C.3 Academic References & Foundational Literature

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